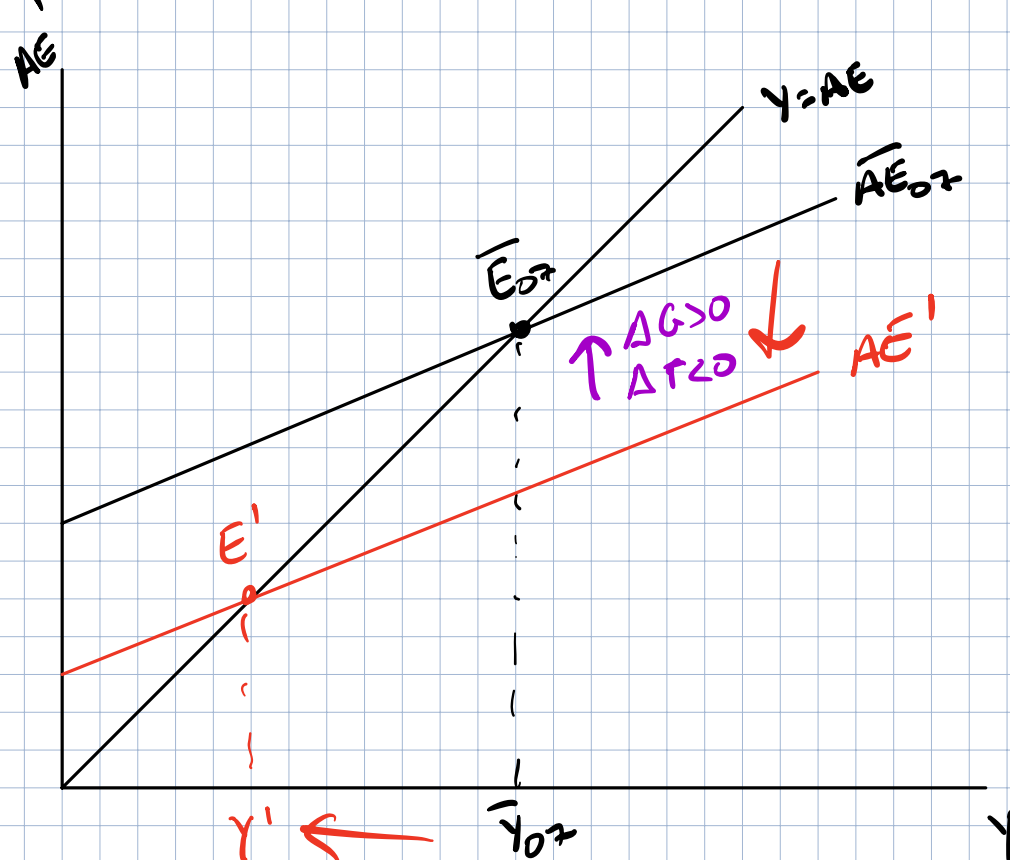


Econ 2 - Lecture 13

Lecture Quiz #6 Released Today, Due Monday, 5/18

Last Class: Fiscal Policy

Recap:



Start at $\bar{Y}$

Crash
 $\bar{AE} \rightarrow AE'$

Gov't can
 $\Delta G > 0$
or $\Delta T < 0$

Cyclical UE

10% UE $\rightarrow$ Gov't Responds

Great Financial Crisis

Bailout Bill $\rightarrow$ Equivalent to a loan

Stimulus Package $\rightarrow$ ARRA

$\Delta G = 500$ Billion, $\Delta T = -300$ Billion

What is expected impact on GDP (Y)?

Consumers during 2008 $\rightarrow$ Low $mpc = 0.50$

$$\text{Expenditure Multiplier} = \frac{1}{1 - mpc} = \frac{1}{1 - 0.5} = 2$$

$$\text{Tax Multiplier} = -\frac{mpc}{1 - mpc} = -\frac{0.5}{1 - 0.5} = -\frac{0.5}{0.5} = -1$$

$$\Delta Y = \frac{1}{1 - 0.5} (\Delta G) + \left[-\frac{0.5}{1 - 0.5} \right] (\Delta T)$$

$$\Delta Y = 2 \times 500 + (-1)(-300) = 1,300 = 1.3 \text{ Trillion}$$

Estimate Expn. Multiplier during 2009 = 1.0 - 2.5

$$\frac{1}{1 - mpc} = 1 \Rightarrow mpc = 0$$

Tax Multiplier = -0.2

$\rightarrow$ Give a \$100 tax cut $\rightarrow$ GDP increases of \$0.20

Covid Pandemic $\rightarrow$ Fiscal Policy

March / April 2020 = \$2 Trillion	} \$5 Trillion $\Delta G, \Delta T$
December 2020 = \$1 Trillion	
March 2021 = \$2 Trillion	

Where does the money come from?

Government Borrowing

$$\text{Deficit} = G - T > 0$$

←

→ Tax Revenue

All Gov't Spending

→ Transfer Payments

→ Interest on Debt

$$2020: G > T \Rightarrow 3 \text{ Trillion}$$

$$2021: G > T \Rightarrow 2.77 T$$

$$2022: \Rightarrow 1.4 T$$

$$2023: \Rightarrow 1.7 T$$

Normalize
Deficit over
time

$$\frac{\text{Deficit}}{\text{GDP}}$$

Today: $\frac{\text{Deficit}}{\text{GDP}} \approx 6\% \rightarrow$ Typically only
greater than 3%
during wars

$$\text{Deficits} \rightarrow \text{Debt} = \text{Sum of Deficits} = \sum (G - T)$$

3 Yr Example

$$\text{Debt} = (G_1 - T_1) + (G_2 - T_2) + (G_3 - T_3)$$

$$= 50$$

$$= 100$$

$$= -75$$

↓

↓

↓

Borrow \$50

Borrow \$100

Surplus = \$75

$$\text{Debt} = 150$$

$$\text{Debt} = 150 - 75 = \$75$$

Current Debt = \$39.24 Trillion

Facts About National Debt

Fact #1: US Government DOES NOT owe
\$39.24 T to outside lenders

Total Debt = Public Debt + Intra-Governmental


Borrow from
outside the
Gov't

= 31.1 T


Borrowing within
the Gov't
→ Social Security,
Dept. of Defense,
etc

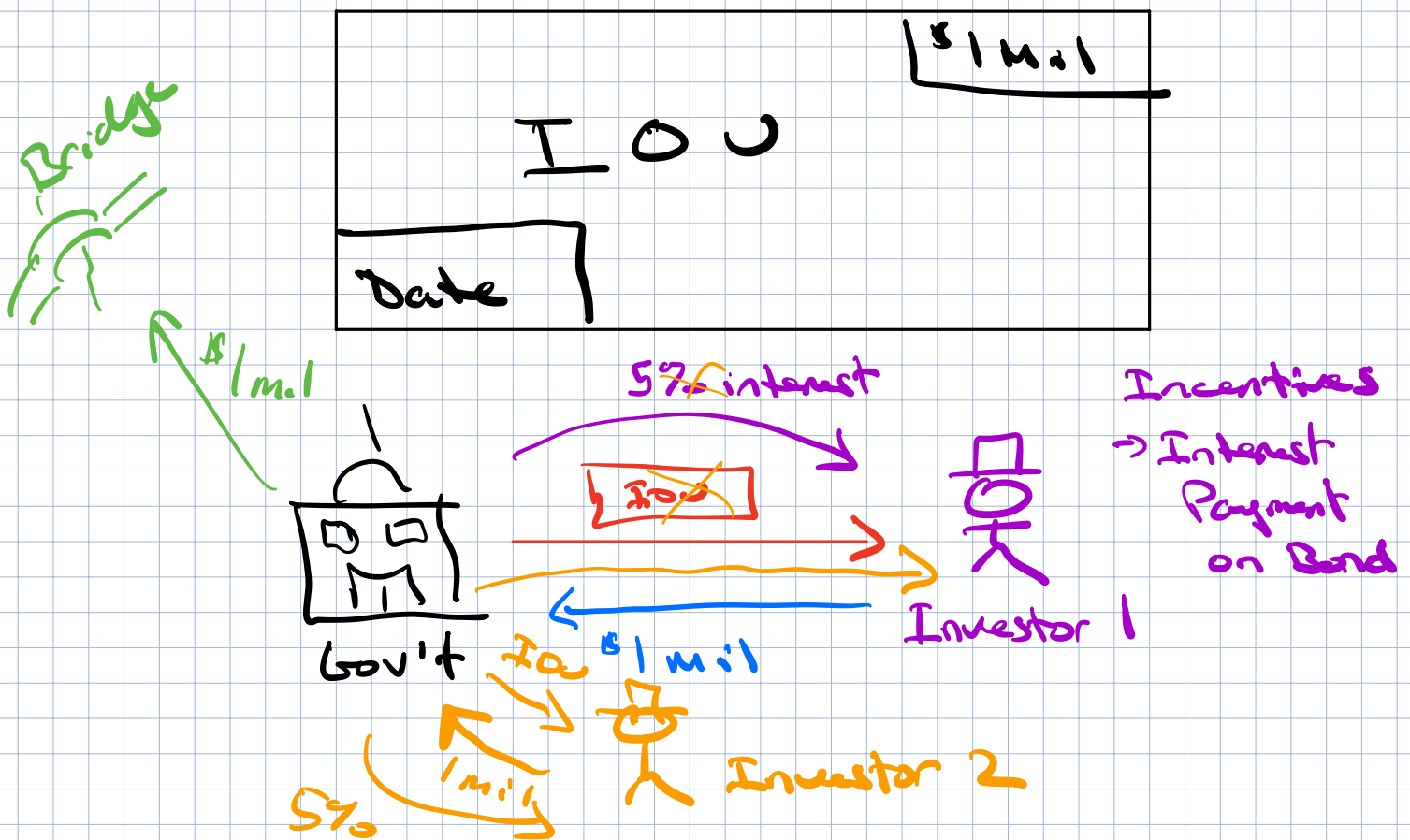
= 8.1 T

Fact #2: We Do NOT need to payback
the 31.1 Trillion in Public Debt

Setting: Gov't borrow \$1 million from lenders

Government Bonds: Vehicle for gov't to
borrow money

"I Owe You" (IOU) $\Rightarrow$ 1 million in one year



In one-year, investor wants \$1 mil back.
Gov't spent \$1 million $\rightarrow$ no money left
Roll over debt to a new lender

$\rightarrow$ Is there a new lender available?

$\rightarrow$ What do we owe on an annual basis?
Interest Payment

What does Gov't need to worry about

Interest Payments on Debt

→ Where does this come from? Tax Payers

Public Debt = \$1 mil

Interest Rate = 5% = 0.05

Nominal GDP = 5 million (Nominal Take-Home income)

$$\begin{aligned}\text{Interest Payment on Debt} &= \text{Interest Rate} \times \text{Debt} \\ &= 0.05 \times 1 \text{ mil} = \$50,000\end{aligned}$$

As long as gov't can pay \$50,000 can hold
1 million in debt → Tax Revenue

$$\underbrace{\text{Tax Revenue}} \geq \text{Interest Payment}$$

$$\text{Tax Rate} \times \text{Nom. GDP} = \text{Interest Payment}$$

$$\text{Minimum Tax Rate} = \frac{\text{Interest Payment}}{\text{Nom. GDP}} \times 100$$

$$\text{Min. Tax Rate} = \frac{50,000}{5,000,000} \times 100 = 1\%$$

⇒ Tax Burden from Debt is 1%

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Public Debt	100,000	200,000
Interest Rate	5%	5%
Price Level	100	120

$$\begin{aligned} \text{Min. Tax Rate} &= \frac{\text{Interest Payment}}{\text{Nom. GDP}} \times 100 \\ \text{Yr 1} &= \frac{100,000 \times 5\%}{200,000} \times 100 = 2.5\% \end{aligned}$$

A 2.5% tax is needed to hold 100,000 in debt

Tax
Burden
is up!

$$\begin{aligned} \text{Min Tax Rate} &= \frac{200,000 \times 5\%}{300,000} \times 100 = 3.33\% \\ \text{Yr 2} & \end{aligned}$$

What is the level of Real GDP in year 2?

	Yr 1	Yr 2
Nom. GDP	200,000	300,000
Price Level	100	120
Real GDP	200,000	→ 250,000

$$\underline{\text{GDP Deflator (Price Index)}} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

$$\text{Real GDP} = \frac{\text{Nom. GDP}}{\text{Price Level}} \times 100$$

$$= \frac{300,000}{120} \times 100 = 250,000$$